

Customer Churn & Retention Analytics with AI

IBM Telco Customer Churn Dataset — End-to-End Machine Learning Project

Author: Kadeejath Muhsina

Technology: Python 3.13.7 | scikit-learn | XGBoost | SHAP | Power BI

1. Introduction

Customer churn — the loss of subscribers or clients — is one of the most costly challenges facing telecommunications companies. Industry research consistently shows that retaining an existing customer is five to seven times less expensive than acquiring a new one. Proactive identification of customers likely to churn, combined with targeted retention interventions, can meaningfully improve both revenue and customer lifetime value.

This project builds a complete, production-quality churn analytics pipeline using the IBM Telco Customer Churn dataset. The pipeline covers every stage from raw data ingestion through machine learning model training, explainability analysis with SHAP (SHapley Additive exPlanations), customer risk segmentation, personalised retention recommendations, and a Power BI-ready dashboard export.

2. Objectives

The project addresses the following objectives: (1) Understand and clean the IBM Telco Customer Churn dataset (7,043 customers, 21 features). (2) Perform exploratory data analysis to uncover churn patterns across contract types, internet services, payment methods, and tenure cohorts. (3) Engineer six domain-specific features to improve predictive performance. (4) Train and compare three classification models: Logistic Regression, Random Forest, and XGBoost. (5) Select and optimise the best model using RandomizedSearchCV. (6) Identify the optimal classification threshold by sweeping precision, recall, and F1 across thresholds 0.05 to 0.95. (7) Predict churn probability for all 7,043 customers and segment them into Low, Medium, and High Risk categories. (8) Apply SHAP to explain global feature importance and individual customer decisions. (9) Generate personalised retention recommendations ranked by business priority score. (10) Produce a Power BI-ready Excel workbook with 8 formatted analytical sheets.

3. Dataset

3.1 Source

The IBM Telco Customer Churn dataset is a widely-used benchmark dataset for churn prediction research. It was originally published by IBM as a sample dataset for their Watson Analytics platform and is available as a publicly downloadable CSV file from IBM's GitHub repository for telco-customer-churn-on-ICP4D.

3.2 Dataset Overview

The dataset contains 7,043 customer records and 21 columns representing customer demographics, subscribed services, account information, and a binary churn label. The class distribution is: 5,174 retained customers (73.5%) and 1,869 churned customers (26.5%), giving an overall churn rate of 26.5%.

3.3 Feature Descriptions

The 21 original features span four categories: (1) Demographics — gender, SeniorCitizen, Partner, Dependents; (2) Account — tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges; (3) Services — PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies; (4) Target — Churn (Yes/No).

3.4 Data Quality Issues

One data quality issue was identified: TotalCharges was stored as object dtype with whitespace entries for 11 new customers whose tenure equals zero. These were converted to numeric (NaN) and imputed with the column median (\$1,397.47). The target column Churn was encoded from Yes/No to 1/0. No other missing values or duplicate records were found.

4. Exploratory Data Analysis

4.1 Churn Distribution

The dataset exhibits a moderate class imbalance with 26.5% positive (churn) cases. This imbalance was addressed in modelling using `class_weight='balanced'` for Logistic Regression and Random Forest, and `scale_pos_weight=2.77` for XGBoost.

4.2 Numeric Feature Analysis

Churned customers show markedly different distributions across all three numeric features compared to retained customers. Churned customers have substantially lower tenure (median approximately 10 months vs 38 months for retained), higher monthly charges, and lower total charges reflecting their shorter tenure. This confirms that early-tenure, high-spend customers are disproportionately likely to churn.

4.3 Categorical Feature Analysis

Contract type is the single most differentiating categorical feature. Month-to-month customers churn at 42.7%, compared to 11.3% for one-year contracts and only 2.8% for two-year contracts. Among internet service types, Fiber optic customers have the highest churn rate at 41.9% versus 19.0% for DSL and 7.4% for customers without internet. Electronic check is the highest-risk payment method at 45.3% churn, compared to 15-19% for automatic payment methods.

4.4 Tenure Cohort Analysis

Churn is highest in the first 12 months (47.4%) and declines consistently with tenure: 28.7% at 13-24 months, 21.6% at 25-36 months, 19.0% at 37-48 months, 14.4% at 49-60 months,

and only 6.6% at 61-72 months. This cohort analysis confirms that early-lifecycle engagement is the most critical retention window.

5. Feature Engineering & Preprocessing

5.1 Encoding

Binary columns (Yes/No, Male/Female) were mapped to 1/0. Multi-category columns (Contract, InternetService, PaymentMethod, MultipleLines, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies) were one-hot encoded with `drop_first=True` to avoid multicollinearity. The final feature matrix contains 36 columns.

5.2 Derived Features

Six domain-specific features were engineered: (1) `avg_monthly_spend` = TotalCharges divided by tenure; (2) `charge_per_month_ratio` = MonthlyCharges divided by (tenure + 1); (3) `num_services` = count of active add-on subscriptions; (4) `tenure_band` = ordinal bucket 0-5 representing 12-month tenure intervals; (5) `is_high_value` = binary flag for customers with charges at or above the 75th percentile (\$79.65); (6) `is_long_term` = binary flag for customers with tenure of 24 months or more.

5.3 Train/Test Split and Scaling

The data was split 80/20 into training (5,634 rows) and test (1,409 rows) sets using stratified sampling to preserve the churn rate in both splits. A `StandardScaler` was fitted exclusively on the training data and applied to both splits, preventing any data leakage from the test set into the scaling parameters.

6. Model Training

Three classification algorithms were trained and evaluated using 5-fold stratified cross-validation on the training set, followed by final evaluation on the held-out test set. Logistic Regression used L2 regularisation ($C=1.0$), `class_weight='balanced'`, `lbfgs` solver, and `max_iter=1000` (training time: 0.08 seconds). Random Forest used 300 estimators, `max_depth=8`, `min_samples_leaf=10`, and `class_weight='balanced'` (training time: 1.22 seconds). XGBoost used 300 estimators, `max_depth=5`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`, and `scale_pos_weight=2.77` (training time: 1.05 seconds).

7. Model Evaluation

7.1 Evaluation Metrics

The primary evaluation metric is ROC-AUC, which is threshold-independent and robust to class imbalance. PR-AUC is additionally reported. F1-score, Precision, Recall, and Accuracy are reported at the optimal threshold.

7.2 Test-Set Results

Logistic Regression achieved the best ROC-AUC (0.8461) and PR-AUC (0.6664). Random Forest achieved the highest F1 (0.6265) and Recall (0.7914). XGBoost achieved the highest Precision (0.5261) and Accuracy (0.7544).

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Accuracy
Logistic Regression (Best)	0.8461	0.6664	0.6138	0.5079	0.7754	0.7410
Random Forest	0.8434	0.6552	0.6265	0.5184	0.7914	0.7495
XGBoost	0.8363	0.6418	0.6198	0.5261	0.7540	0.7544

Table 1: Model performance on held-out test set (n=1,409 customers, 374 churned)

7.3 Cross-Validation Results

5-fold cross-validation confirmed stable performance. Logistic Regression CV ROC-AUC: 0.848 +/- 0.011. Random Forest: 0.846 +/- 0.012. XGBoost: 0.842 +/- 0.010. Low standard deviations indicate no overfitting.

8. Model Optimisation

Logistic Regression (best by ROC-AUC) was further optimised using RandomizedSearchCV with 40 iterations and 5-fold stratified CV, scoring by ROC-AUC. Search space: C in {0.001, 0.01, 0.1, 1, 10, 100}, penalty in {L1, L2}, solver in {liblinear, saga}. Best parameters: C=0.1, penalty=L1, solver=liblinear. Best CV ROC-AUC: 0.8485. Optimised model test-set: ROC-AUC=0.8452, PR-AUC=0.6628, F1=0.6183, Recall=0.7754.

9. Threshold Analysis

A systematic threshold sweep from 0.05 to 0.95 (step 0.01) was performed, recording Precision, Recall, F1, and Accuracy at each point. The optimal threshold was defined as the value that maximises F1. Optimal threshold: 0.57. At t=0.57: Precision=0.544, Recall=0.741, F1=0.627, Accuracy=0.767. This represents a 1.5 pp improvement in F1 and a 2.7 pp improvement in Accuracy compared to the default threshold of 0.50.

10. Churn Probability Prediction & Risk Segmentation

The optimised model was applied to all 7,043 customers. Customers were segmented using fixed probability thresholds: Low Risk (probability below 30%), Medium Risk (30-60%), High Risk (60% and above).

Segment	Probability	Count	% of Total	Actual Churn Rate
Low Risk	Below 30%	2,916	41.4%	5.5%
Medium Risk	30% - 60%	912	12.9%	19.4%
High Risk	60% and above	3,215	45.6%	47.7%

Table 2: Customer risk segmentation results

High-risk customers (3,215) have an average churn probability of 88.3% and an actual churn rate of 47.7%. Low-risk customers have a mean churn probability of 8.2% and an actual churn rate of 5.5%.

11. SHAP Explainability

11.1 Method

SHAP (SHapley Additive exPlanations) provides theoretically grounded, model-agnostic feature attributions based on Shapley values from cooperative game theory (Lundberg &

Lee, 2017). For Logistic Regression, a LinearExplainer was used. SHAP values were computed for a stratified sample of 500 test observations.

11.2 Global Feature Importance

The top 10 features by mean absolute SHAP value are: (1) Contract_Two year (1.304) — the most powerful protective factor; (2) InternetService_Fiber optic (0.747) — strongest positive risk factor; (3) Contract_One year (0.728) — second protective factor; (4) PaymentMethod_Electronic check (0.331) — payment friction risk signal; (5) OnlineSecurity_Yes (0.328) — service stickiness protective factor; (6) PaperlessBilling (0.310); (7) MultipleLines_Yes (0.272); (8) InternetService_No (0.270) — protective; (9) StreamingMovies_Yes (0.259); (10) StreamingTV_Yes (0.229).

11.3 Individual Explanations

SHAP waterfall plots were generated for the three highest-probability churners in the test set. These plots show the contribution of each feature to moving the prediction from the base rate to the individual customer's predicted churn probability, enabling customer success teams to understand precisely why a specific customer is flagged.

12. Retention Recommendations

A rules-based recommendation engine evaluates 11 signal categories per customer and assigns up to three prioritised retention actions. Customers are ranked by Priority Score = churn_probability multiplied by (monthly_charges multiplied by (72 - tenure)) divided by 100. This weights customers by both churn risk and remaining lifetime value.

A total of 4,127 at-risk customers received personalised recommendations. Top five retention strategies: (1) Contract upgrade offer — discounted 1-year contract for month-to-month customers; (2) Online security bundle at 30% introductory discount; (3) Free 3-month tech support trial; (4) Auto-pay migration incentive (\$5/month bill credit); (5) New customer onboarding specialist assignment.

13. Power BI Dashboard

A Power BI-ready Excel workbook was produced with 8 formatted analytical sheets: Customer_Overview (7,043 rows), Risk_Segments (segment KPIs), Churn_Drivers (SHAP importance for 36 features), Retention_Actions (4,127 customers with recommended actions), Model_Performance (3-model comparison), Tenure_Cohort_Analysis (6 cohorts), Service_Churn_Rate (26 service breakdowns), and Monthly_Charges_Bins (6 charge brackets). All sheets use colour-coded risk rows, frozen headers, and consistent formatting for direct Power BI import.

14. Business Insights

Five actionable business insights emerge from this analysis: (1) Contract commitment is the strongest retention lever — two-year contracts have a 2.8% churn rate versus 42.7% for month-to-month, a 15-fold difference. (2) The first 12 months are the critical retention window — 47.4% churn in year one declines to 6.6% after year five. (3) Fiber optic customers face a value gap at 41.9% churn, requiring targeted satisfaction outreach. (4)

Payment method is a strong churn signal — electronic check payers churn at 45.3% versus 15.2% for credit card auto-pay. (5) Annual revenue at risk is approximately \$2.5 million — a 20% retention rate among high-risk customers protects approximately \$500,000 per year.

15. Limitations

(1) Static snapshot: The dataset represents a single point in time. A survival model would better capture time-to-churn dynamics. (2) No external data: Customer satisfaction scores, support ticket history, and competitor pricing are absent. (3) Class imbalance limits precision for the minority class despite balancing strategies. (4) SHAP values were computed on a 500-sample subset for computational efficiency. (5) The rules-based recommendation engine is manually designed; uplift modelling would produce more validated intervention strategies.

16. Future Work

(1) Deploy the model as a REST API (FastAPI or Flask) for real-time churn scoring. (2) Implement a survival analysis model to predict time-to-churn. (3) Build an uplift model to identify customers who respond positively to retention interventions. (4) Connect the Power BI workbook to a live database with automated weekly model retraining. (5) Extend SHAP explanations to individual customer-level CRM cards for customer success managers.

17. Conclusion

This project demonstrates a complete, end-to-end machine learning pipeline for customer churn prediction and retention strategy development. Using the IBM Telco Customer Churn dataset, three classification models were trained, evaluated, and compared. Logistic Regression with L1 regularisation was selected as the best model, achieving a ROC-AUC of 0.8461 and identifying 77.5% of churning customers at the optimal threshold of 0.57.

SHAP analysis confirmed that contract type, internet service type, payment method, and online security subscriptions are the primary drivers of churn — findings with direct, actionable implications for product and retention teams. The risk segmentation identified 3,215 high-risk customers with an average churn probability of 88.3%, representing approximately \$2.5 million in annual revenue at risk. The project delivers not only a predictive model but a complete analytics package: personalised retention recommendations for 4,127 at-risk customers, a formatted Power BI workbook, and a fully reproducible Python codebase with modular architecture suitable for production deployment.

References

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